Question:

A 27-year-old man presents to the emergency department with severe substernal pain at rest, which radiates to his left arm and jaw. He reports that he has had similar but milder pain several times in the past during strenuous exercise. He had heart transplantation due to dilatory cardiomyopathy 5 years ago with an acute rejection reaction that was successfully treated with corticosteroids. He had been taking 1 mg tacrolimus twice a day for 3.5 years but then discontinued it and had no regular follow-ups. The man does not have a family history of premature coronary artery disease. His blood pressure is 110/60 mm Hg, heart rate is 97/min, respiratory rate is 22/min, and temperature is 37.3°C (99.1°F). On physical examination, the patient is alert, responsive, and agitated. Cardiac auscultation reveals a fourth heart sound (S4) and an irregularly irregular heart rhythm. His ECG shows ST elevation in leads I, II, V5, and V6, and ST depression in leads III and aVF. His complete blood count and lipidogram are within normal limits. The patient's cardiac troponin I and T levels are elevated. A coronary angiogram reveals diffuse concentric narrowing of all branches of the left coronary artery. What is the most likely causative mechanism of this patient's cardiac ischemia?Answer Choices:A. Vasospasm of distal coronary arteries branchesB. Left ventricular hypertrophyC. Obliterative arteriopathyD. Increased oxygen demand due to tachycardia

Reasoning + Answer:

<think>The patient has a history of heart transplantation and is currently showing signs of cardiac ischemia with ST elevation on ECG , which can be indicative of coronary artery disease or vasospasm , especially in a transplanted heart where the native coronary arteries may have been damaged or the transplanted coronary arteries may be at risk for spasm leading to ischemia even without traditional risk factors for coronary artery disease.</think><answer>A . Vasospasm of distal coronary arteries branches</answer>